

Private or Public Equity?

The Evolving Entrepreneurial Finance Landscape

Michael Ewens (Caltech and NBER)

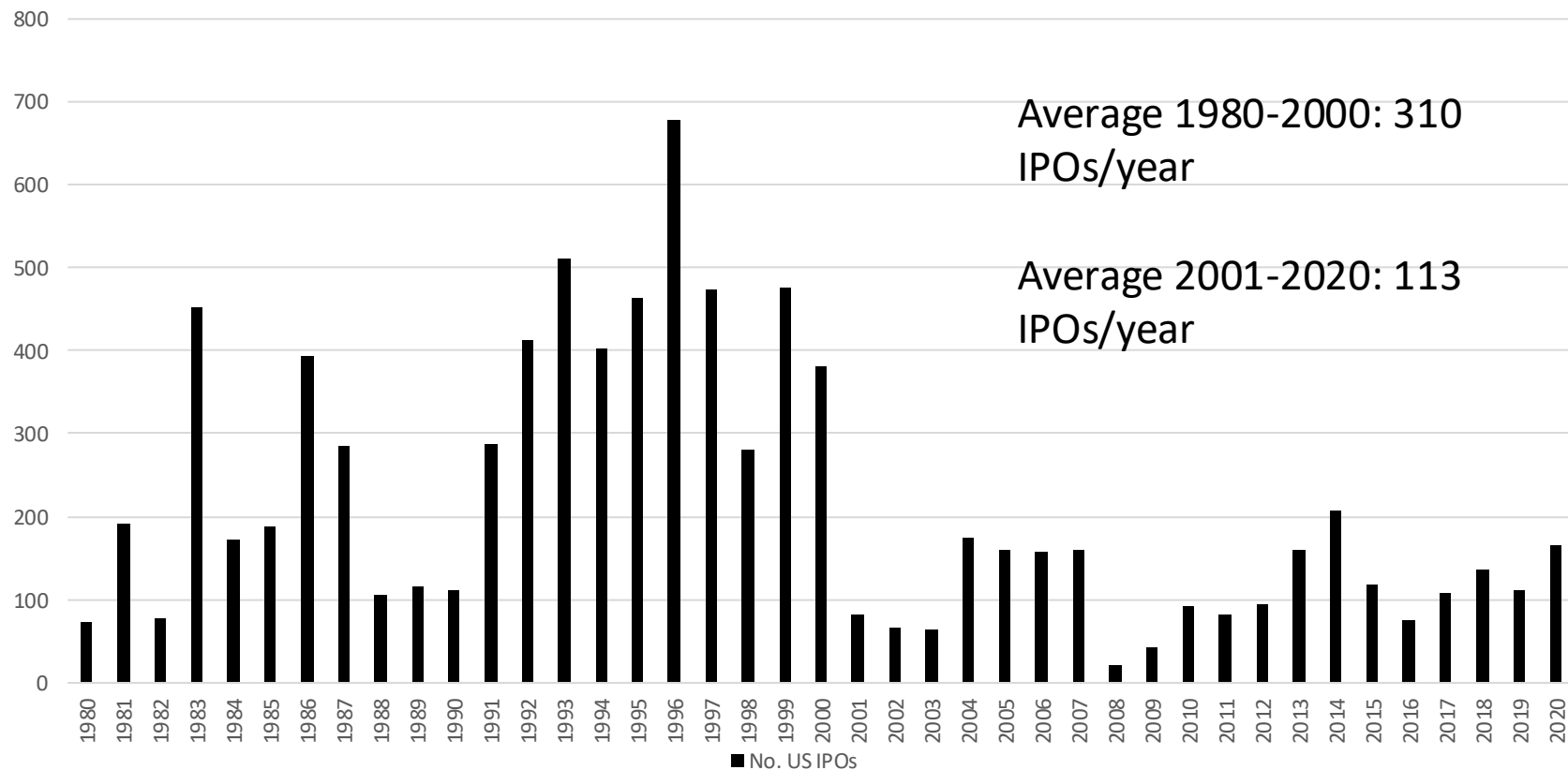
Joan Farre-Mensa (University of Illinois at Chicago)

April 2022

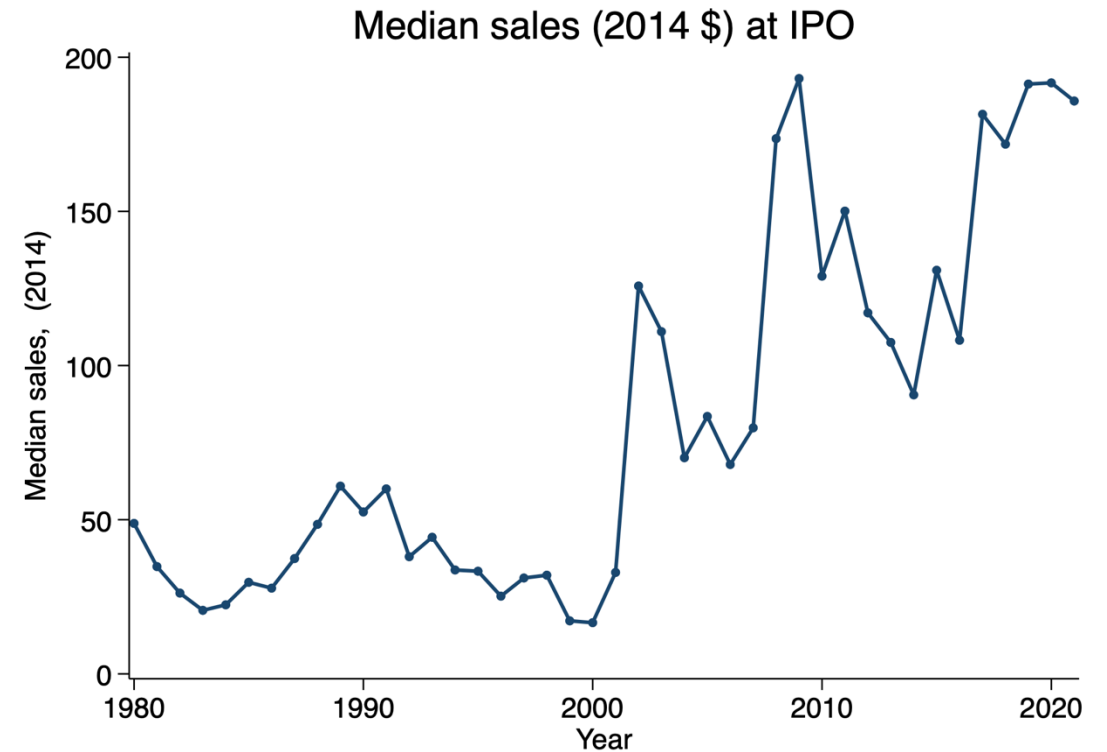
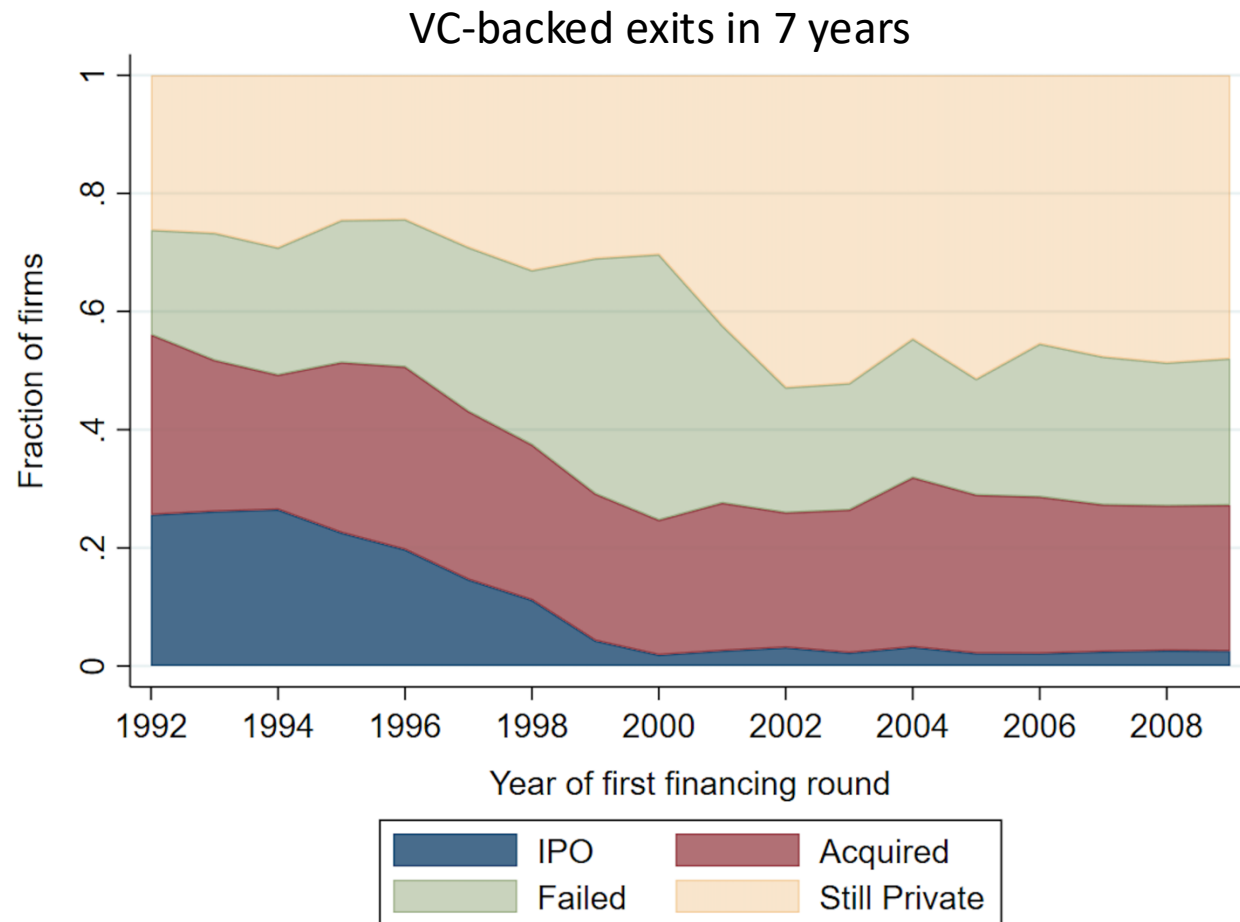
Four motivating facts

Fewer IPOs and publicly-listed firms

Since the 1996, sharp decline in the number of IPOs in the U.S.

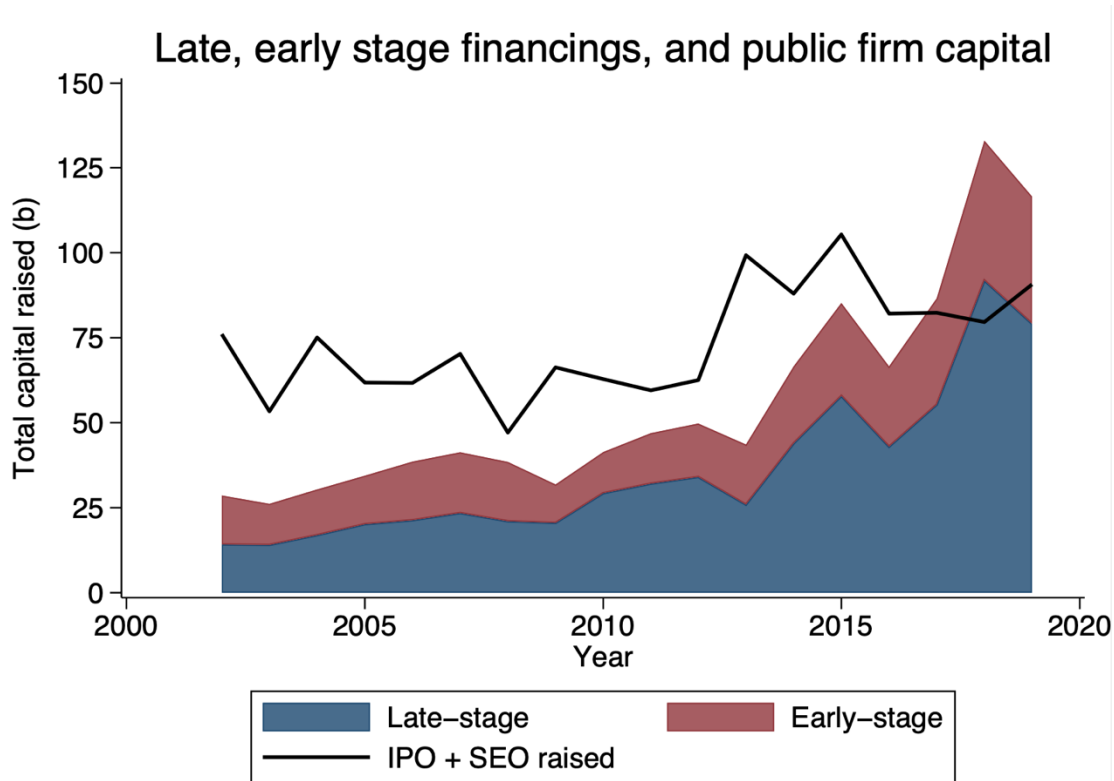


Private longer and larger at IPO

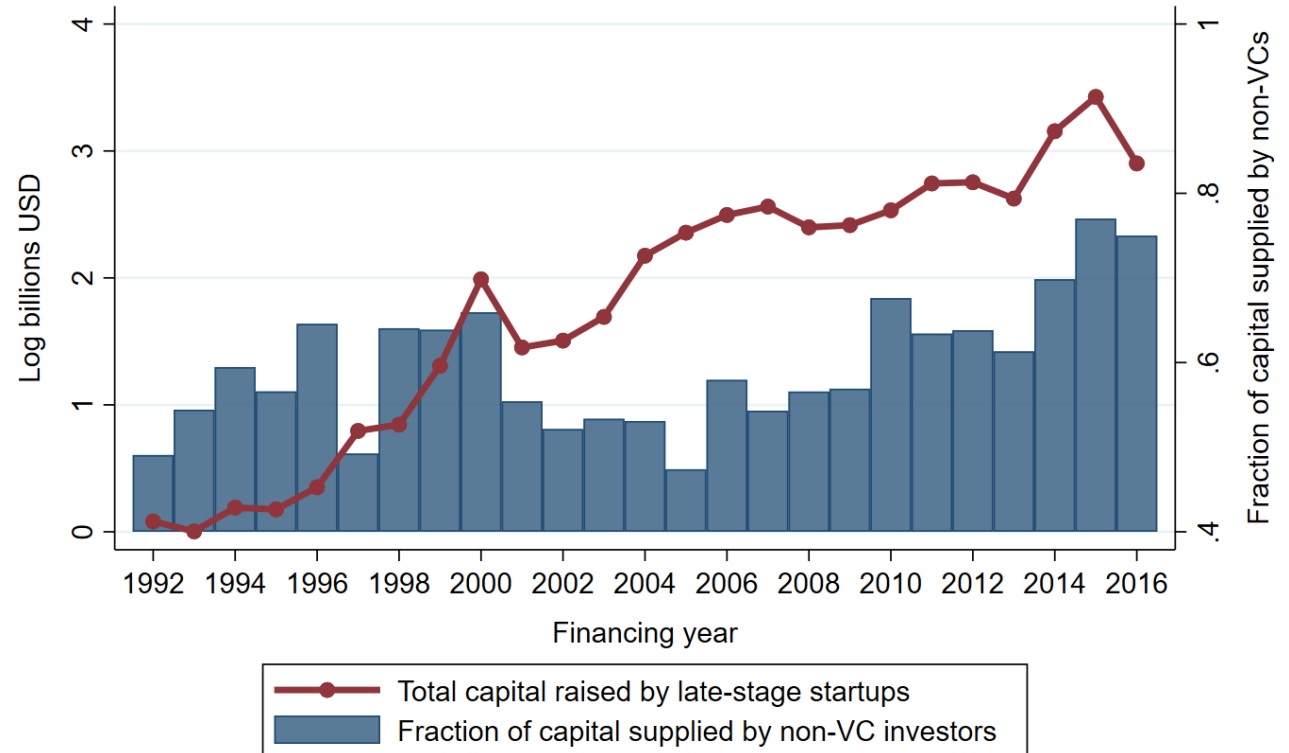


Ritter (2022)

Late-stage startup financings: more capital from new (and old) investors



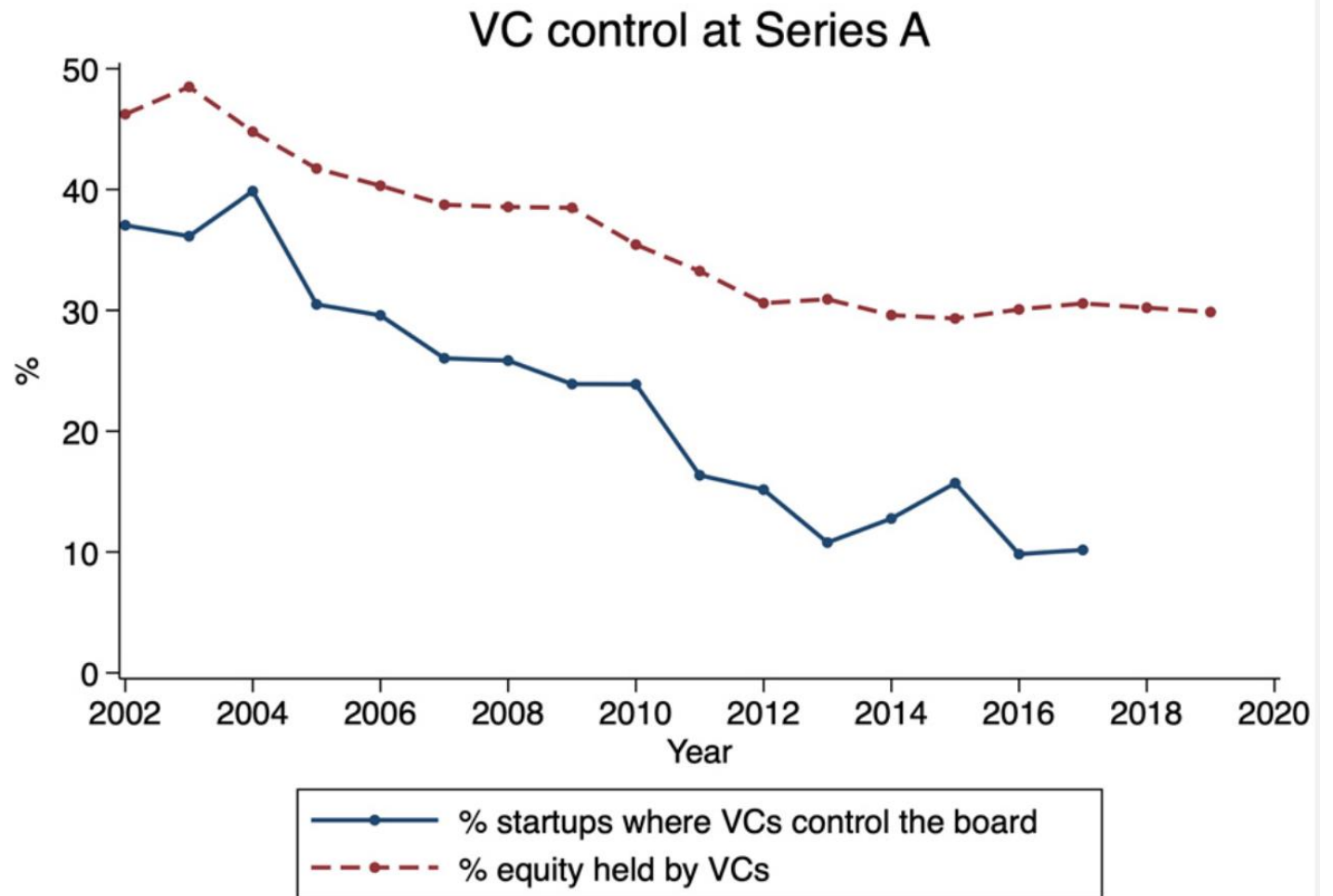
Growth in private capital driven by late-stage financings



Late-stage financings include more non-traditional investors

Early-stage financing: “cheaper” capital

- Startup founders sell less equity in the early stages
- Startup founders more likely to control their boards



Interconnection btw public & private markets

- Significant changes in public and private capital markets over last 25+ years
- Changes are interconnected
 - E.g., private capital regulations affect public-private trade-off
- Entrepreneurs and investors evaluate public vs. private trade-offs differently
 - Changes alter not only the public vs. private trade-off and agents' bargaining power when making exit decisions
- Important for research and policy to account for dynamic interconnection

Questions

- How have public and private markets changed over the last 25 years?
- What are the drivers of these changes?
- How can policy be guided by these new facts?
- Where do researchers need to look next?

Setting the stage

Describing players, differences btw public and private firms

Defining public, private, and semi-public firms

- Public firm: Firm whose shares are listed in a major stock exchange → Shares need to be registered
- Two types of semi-public firms:
 - shares listed in “minor” exchanges (e.g., OTC, BB, pink sheets).
 - have to file financial statements w/ SEC
 - Lots of shareholders or registered securities (e.g., bonds)
- (Pure) private firm= All other firms



Quick private markets primer

- **Early-stage startup:** typically, pre-revenue firms, less than 4 y.o.
- **Late-stage startup:** larger, older startups from 4-10 y.o.
 - Startups likely to go public
- **Security design:** startups raise preferred equity or convertible notes
 - Investors tend to have more control rights than common shareholders
- **Intermediated capital:** Most investors provide capital out of closed-end, non-registered funds (traditionally, VC funds)
 - Goal: return capital to their investors w/in 7-12 years

Raising capital as a private firm

- Private firms issue securities → meet one of the exemptions from registration of Section 4(a)(2) of the Securities Act
- Most common exemption is Rule 506 of Regulation D
 - Limits to whom one can sell, how many and liquidity
- Or, Rule 504, Reg A and Reg crowdfunding → offering size limit
- Securities sold under Rule 506 are restricted securities that cannot be easily traded
 - Secondary trades often require issuer cooperation

From private to public firm

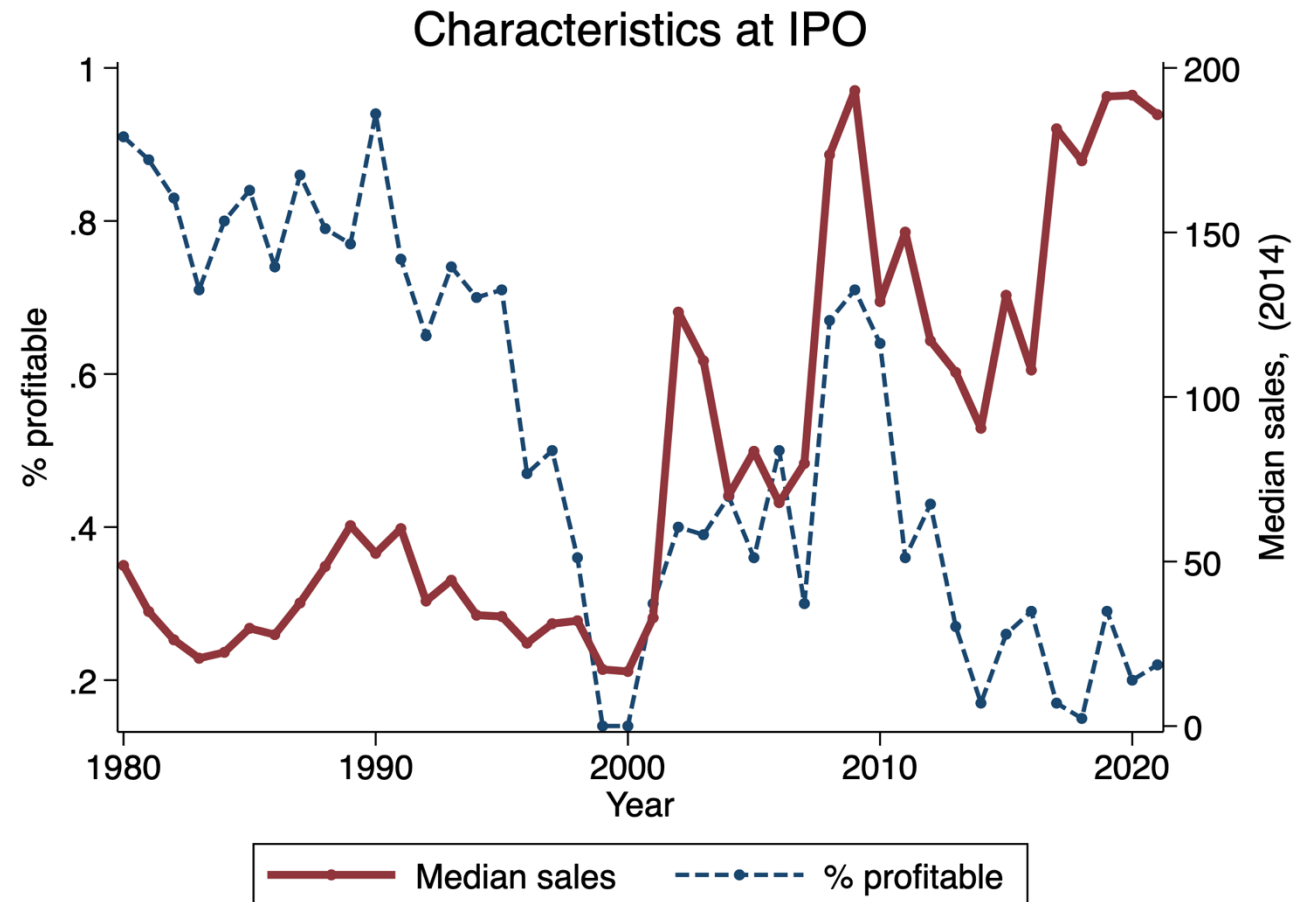
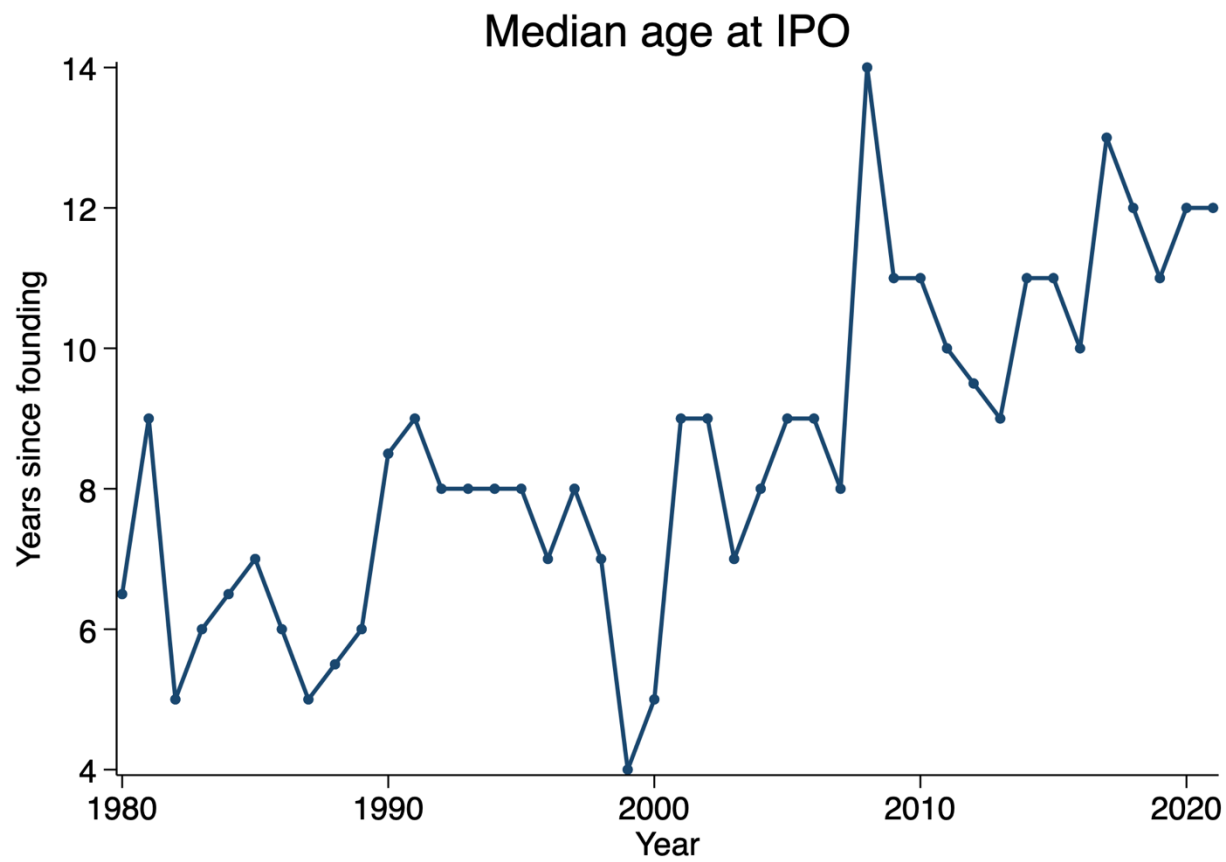
- **Share liquidity** increases → lower cost of capital
 - restrictions on number/type shareholders lifted, solicitation restrictions lifted
- **Regulatory** environment changes
 - Exchange listing requirements + SOX auditing and internal reporting controls
- **Information** environment → two-audiences problem (Bhattacharya & Ritter 1983)
 - file quarterly & annual financial statements
 - subject to insider trading laws and not allowed to selectively disclose material
- Greater **separation ownership & control** → greater scope for agency pbms
 - Short-termist pressures with offset increase in corporate governance rules

Changes to the public capital
markets

Major regulations and public market changes

- Sarbanes-Oxley Act (2002), particularly Section 404 (Coates and Srinivasan, 2014)
 - Decline in “ecosystem” of underwriters that focuses on smaller firms
 - Regulation Fair Disclosure (2000)
 - 2003 Global Settlement
 - JOBS Act (2012)
- Gao et al. (2013), Doidge et al. (2013, 2017): These changes are unlikely to explain decline in IPOs

Older, bigger, and less profitable at IPO



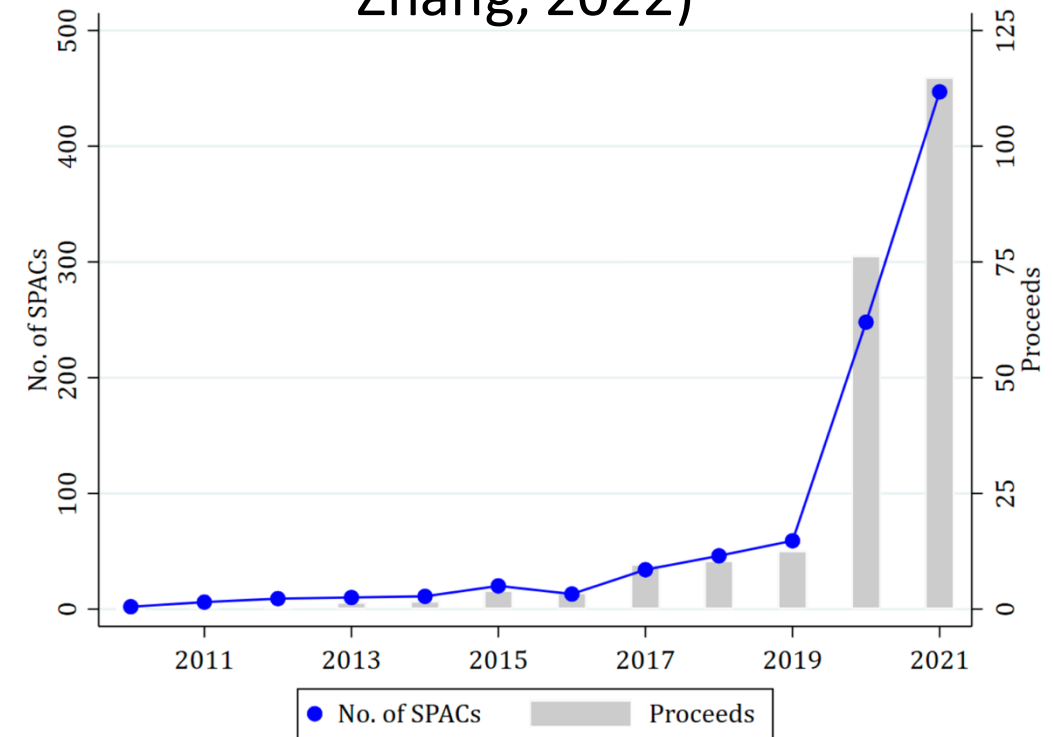
Innovation in listing options

Nasdaq Direct Listings offer a different way to go public with unrestricted liquidity and no lock-up period

Choose Your Path to Public:

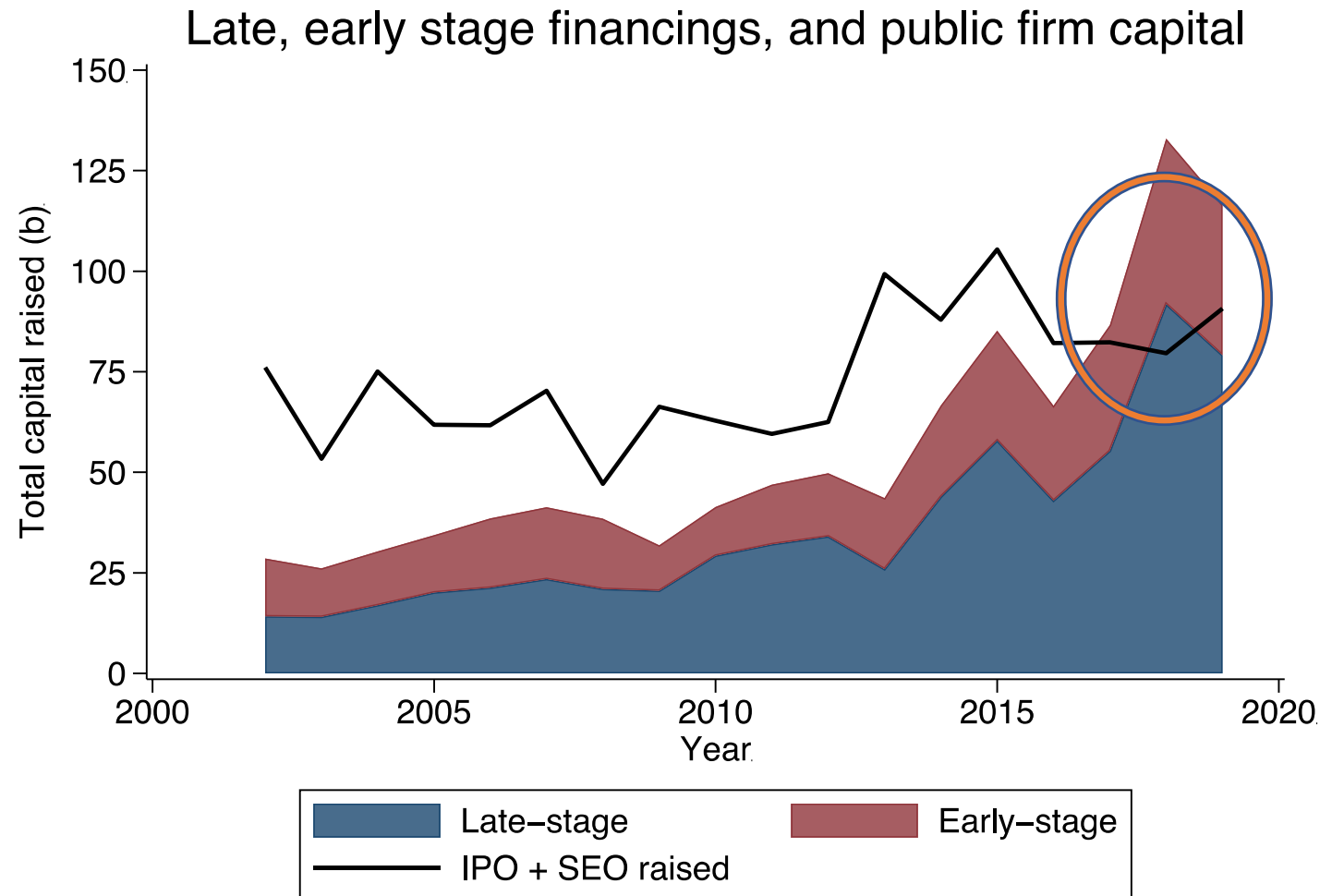
Direct Listing, now with a capital raise

Increase in SPAC activity (Gahng, Ritter and Zhang, 2022)



Public equity capital raising vs. VC

Since 2017, capital raised by VC-backed startups > IPO and SEO offerings

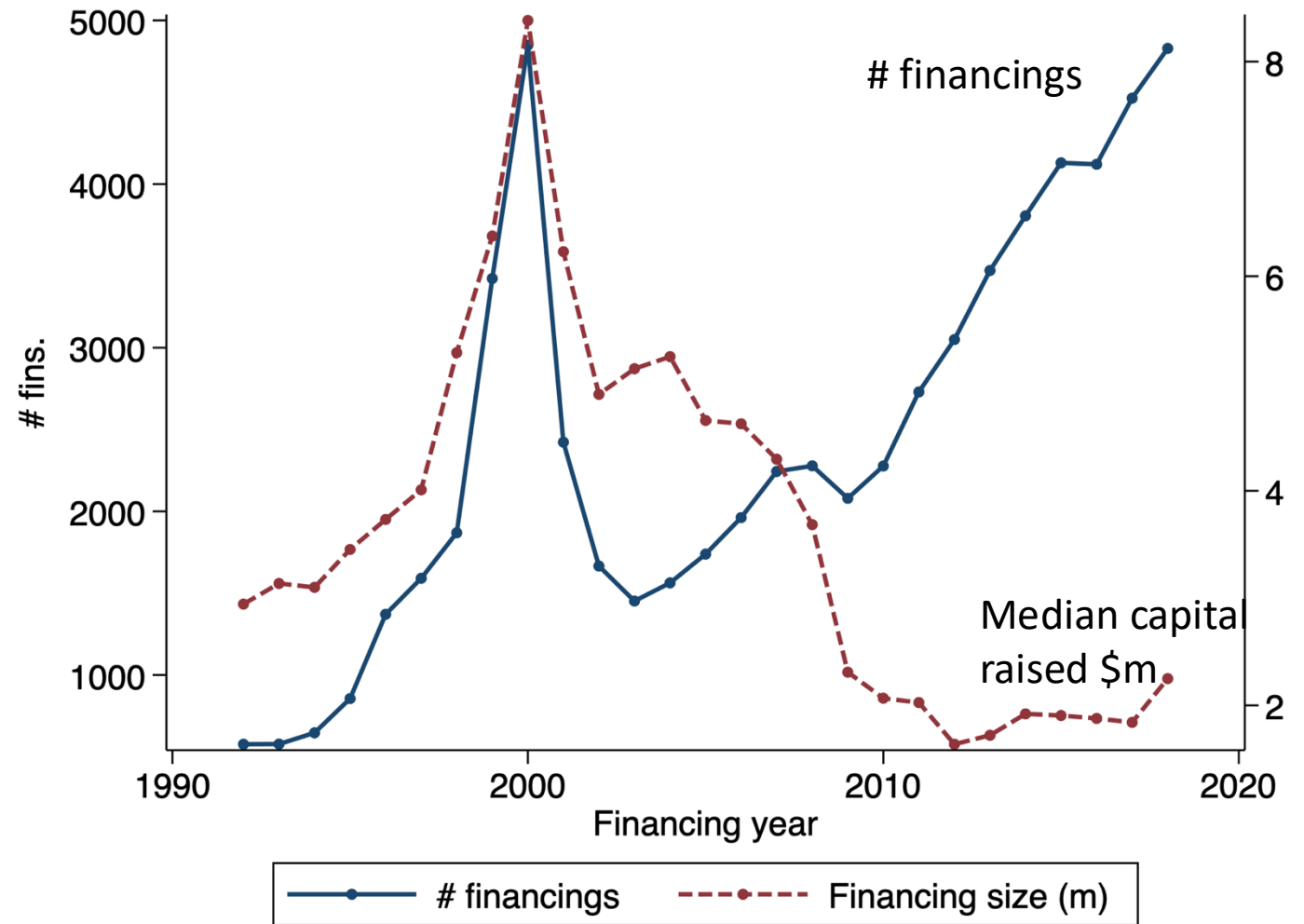


Private capital market changes

More startups and a lot more capital

More early-stage startups raising smaller rounds

- Early-stage startup financings back to dot-com peak
- Median round size fallen 2.5X since 1997



New early-stage investors and changing VC

 AngelList Venture

For Investors ▾

**Invest deal-by-deal in
world-changing
startups**



techstars

Accelerators and ecosystems

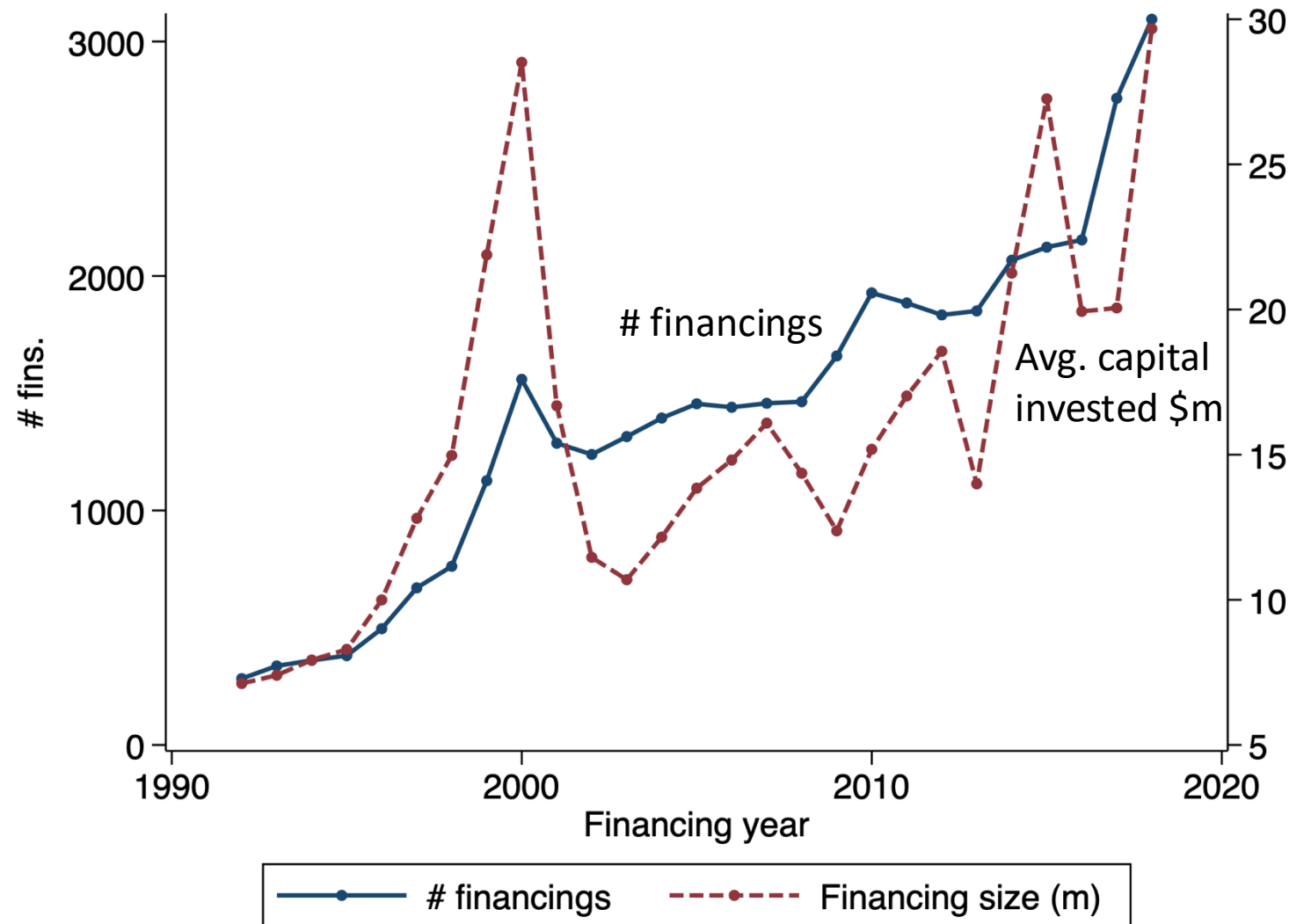
By Yael V. Hochberg^{6,7,21} and Daniel C. Fehder⁷

	2002	2008	2014	2019
Investor pool				
Unique early-stage investors	1,552	1,352	2,029	2,524
Financings with angel investors	n/a	709	3,657	3,125
Unique accelerators/incubators	111	248	787	1078
Equity crowdfunding platforms	0	0	23 [*]	34

Since early 2000s, emergence of new early-stage and retail investors using non-traditional financing model

Late-stage financing growth

- 90th percentile IPO size pre-1997: \$99m
- 2002, 6 financings > \$99m
- 2019: 189 financings > \$99m



Changes in the capital supply since 2002

	2002	2008	2014	2019
Capital supply				
\$b raised by VC funds	\$14.7b	\$27.7b	\$36.7b	\$52.3b
\$b raised by PE funds	\$68.1b	\$202.8b	\$179b	\$304b
Median VC fund size (\$m)	\$61.3m	\$78.6m	\$23.3m	\$31.7m
Public pension funds allocation to VC/PE	2%	4.1%	5.4%	7%
High-ed endowments allocation to VC/PE	5.5%	11.3%	15%	19.9%
VC funds dry powder (\$b)	\$122.6b	\$115b	\$103.5b	\$233b

Roughly 3X increase from 2002 – 2019

- VC funds raised
- VC/PE allocations by public pensions
- VC/PE allocations for higher-ed endowments

What drives these changes?

- Regulation
- Changes affecting investors (supply side)
- Changes affecting startups and founders (demand side)

Regulation (and deregulation)

1. NSMIA (1996)

2. Pension Protection Act (2006)

- Easier for VC/PE funds to raise capital from governmental pension plans w/o triggering costly fiduciary rules

3. JOBS Act (2012) and regulatory follow-ups

- AngelList no-action letter
- Reg Crowdfunding (provides regulatory framework)
- Easing of no-solicitation rules (Rule 506(c))

NSMIA (National Securities Market Improvement Act of 1996)

- Federal preemption of state blue-sky laws for private capital raising
→ easier to raise private capital across state lines, most helpful for late-stage startups
- ↑ cap on # of investors that trigger fund registration
→ VC/PE funds can raise \$ from more investors w/o registering, most helpful for late-stage funds aiming to raise large funds

	Has out-of-state investors?		
	Non-IT		
Late stage $\times$ Post	0.041*** (0.012)	0.070*** (0.02)	
Late stage round	0.173*** (0.028)	0.138*** (0.022)	
Log round # $\times$ Post			0.049*** (0.011)
Log round #			0.176*** (0.026)
Financings	8244	3038	8244
R^2	0.109	0.125	0.136
State / Ind. FE?	Y	Y	Y
Year-quarter FE?	Y	Y	Y

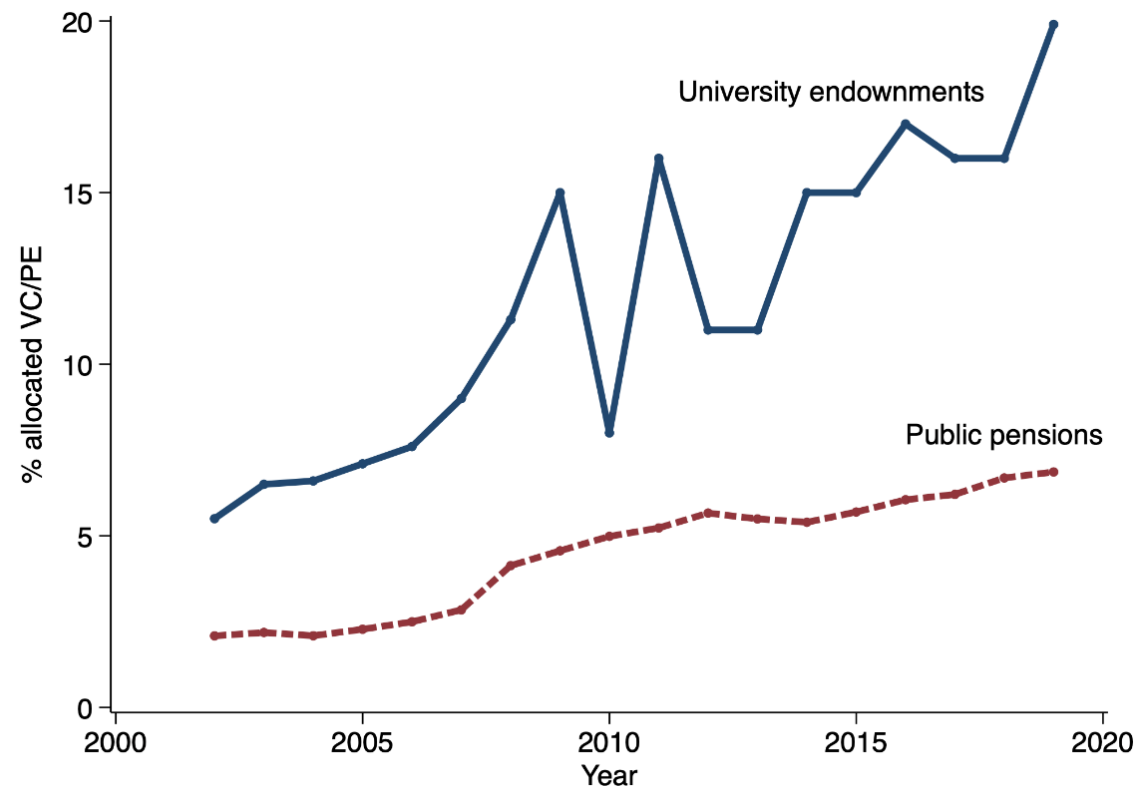
Ewens and Farre-Mensa (2020)

- Late-stage startups (“treated” by NSMIA) increase reliance on out-of-state investors after NSMIA relative to early-stage startups
- Increase driven by startups in states w/ non-standardized blue-sky laws (triple-diff)

- After NSMIA, similar diff-in-diffs show ...
 - late-stage startups able to raise larger amounts of capital
 - late-stage focused funds able to raise larger funds
 - Late-stage startups == IPO candidates

Changes affecting investors (non-regulatory)

1. More capital allocated to VC/PE funds by some of their major LPs, s.a. higher-ed endowments and public pension funds



Changes affecting investors

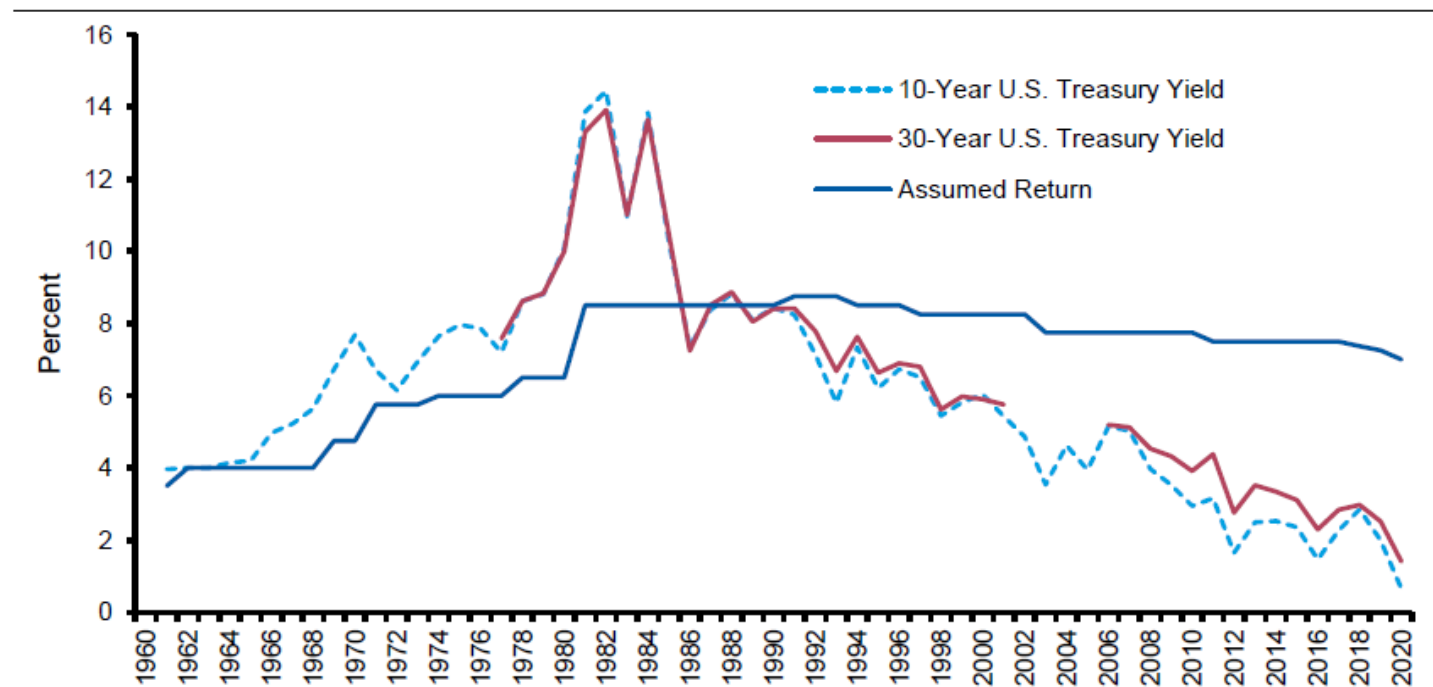
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Why?

- Higher-ed: chasing the Yale endowment returns? (Lerner et al. 2008)
- Public pension funds: Pension gap from \$0 in 2001 to \$1.25 trillion in 2019
 - Higher expected returns → lower gap
 - Public pension funds assign highest expected return VC/PE, mean expected net return of 12% (Andonov & Rauh 2021)
 - Compare to 1.5% Treasury rate in 2020, down from 5.2% in 2001

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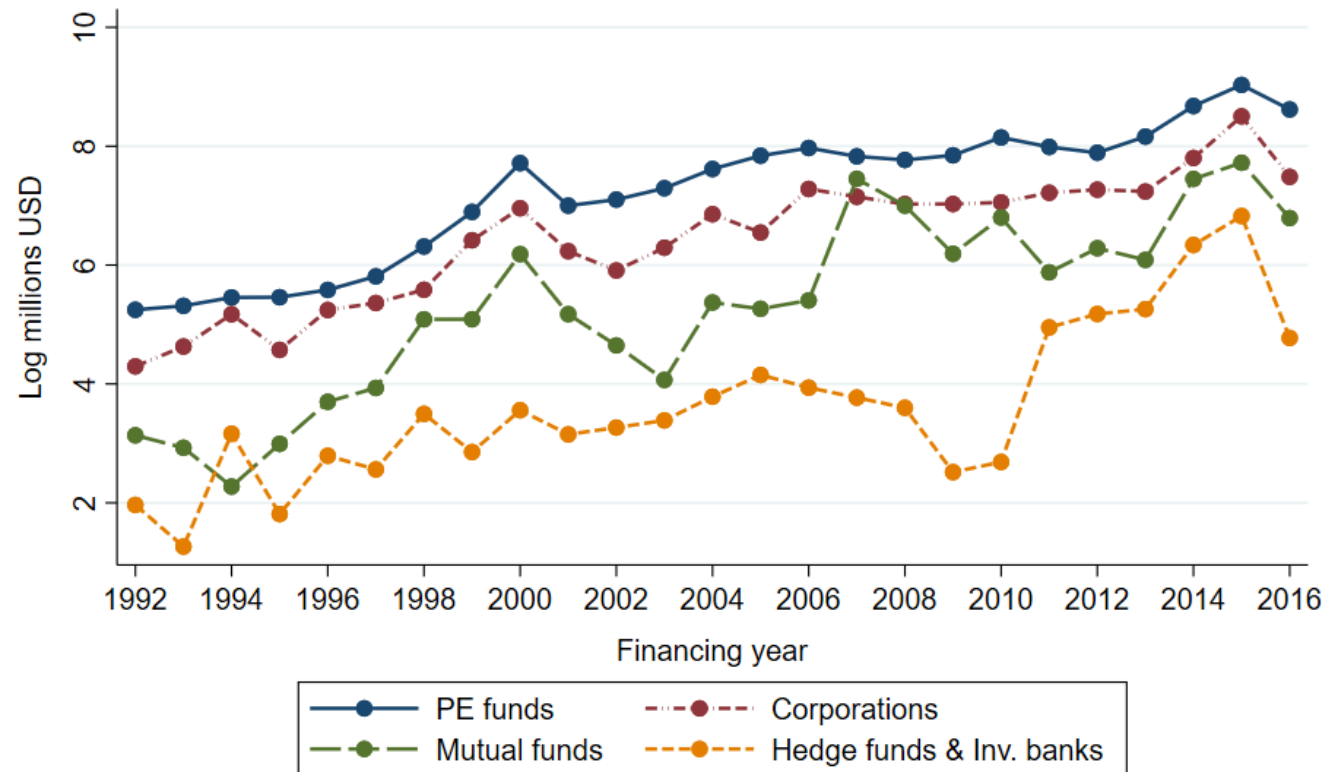


Source: CalPERS; Board of Governors of the Federal Reserve System (U.S.); and FactSet.

Note: Assumed returns and Treasury yields reflect CalPERS's June 30 fiscal year.

Changes affecting investors

2. Increased willingness of traditional public investors to invest in private mkts (Aragon et al. 2018, Kwon et al. 2020, Agarwal et al. 2021, Chernenko et al. 2021, Huang et al. 2021)



Changes affecting investors

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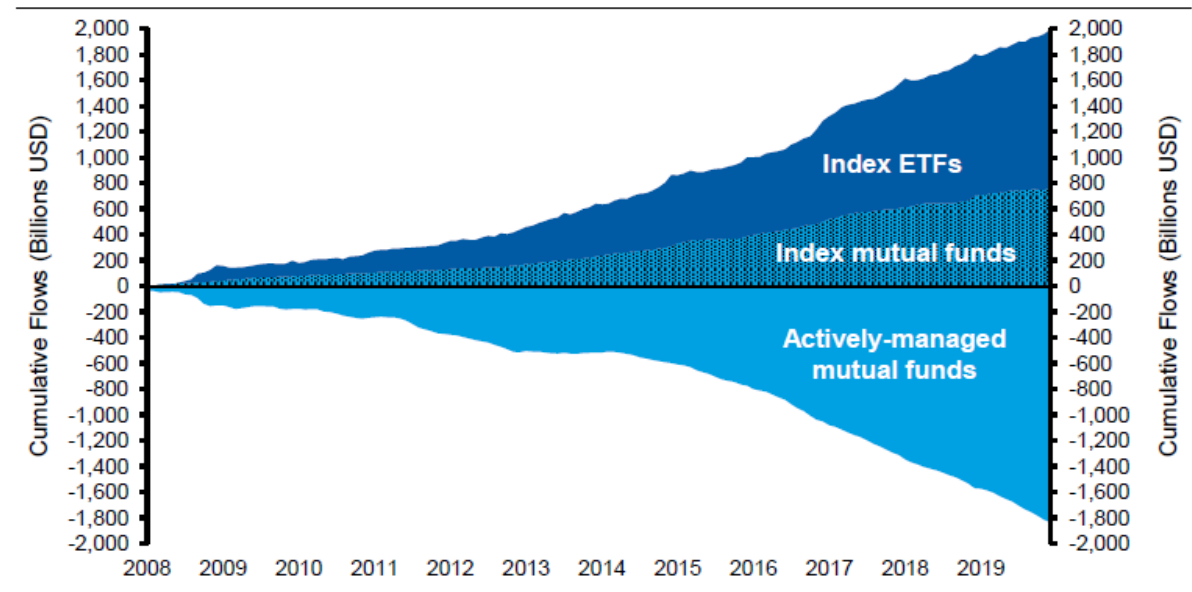
Why?

- Traditional IPO firms are now in private markets (circular argument...)
- Investing in private firms can help mutual/hedge funds beat passively managed index funds and ETFs—and so justify their higher fees

Changes affecting investors

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Investing in private firms can help mutual/hedge funds beat passively managed index funds and ETFs—and so justify their higher fees



Source: Investment Company Institute.

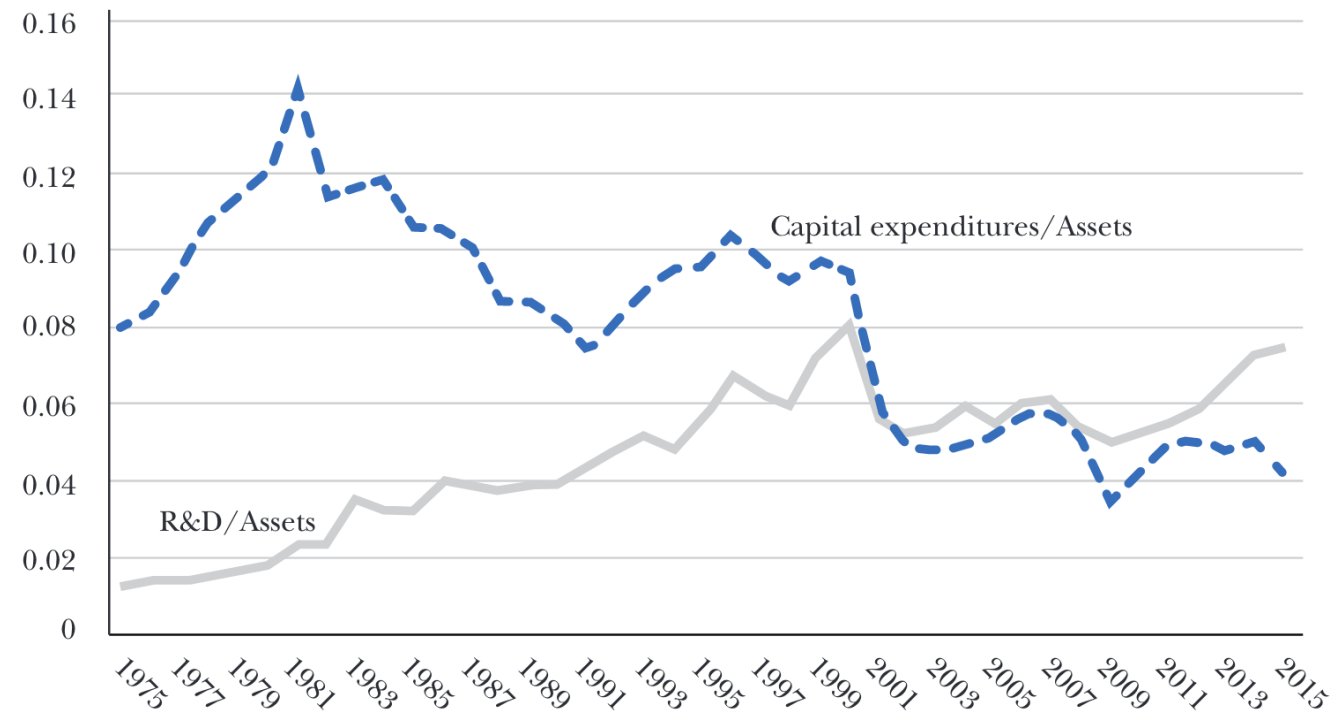
Note: U.S. domestic equity funds; Mutual fund data is net new cash flow plus reinvested dividends; ETF data is net share issuance and includes reinvested dividends.

Changes affecting startups and founders

Kahle and Stulz (2017):

“Raising equity in public markets to fund R&D can be difficult. Investors want to know what they invest in, but the more a firm discloses, the more it becomes at risk of providing ammunition to its competitors. As a result, **R&D-intensive firms may be better off raising equity privately** from investors who then have large stakes.”

Evolution of Capital Expenditures and R&D from 1975 through 2015
(as a ratio of total assets)



Changes affecting startups and founders

2. Technological innovations have decreased startups' capital needs, particularly during the initial, experimental stage of the entrepreneurial process—**when capital is most expensive**

- Cloud computing's emergence in 2006 (AWS)
 - build software products w/o significant upfront investments
 - raise less capital (Ewens et al. 2018)
- Also, biotech (e.g., CRISPR)

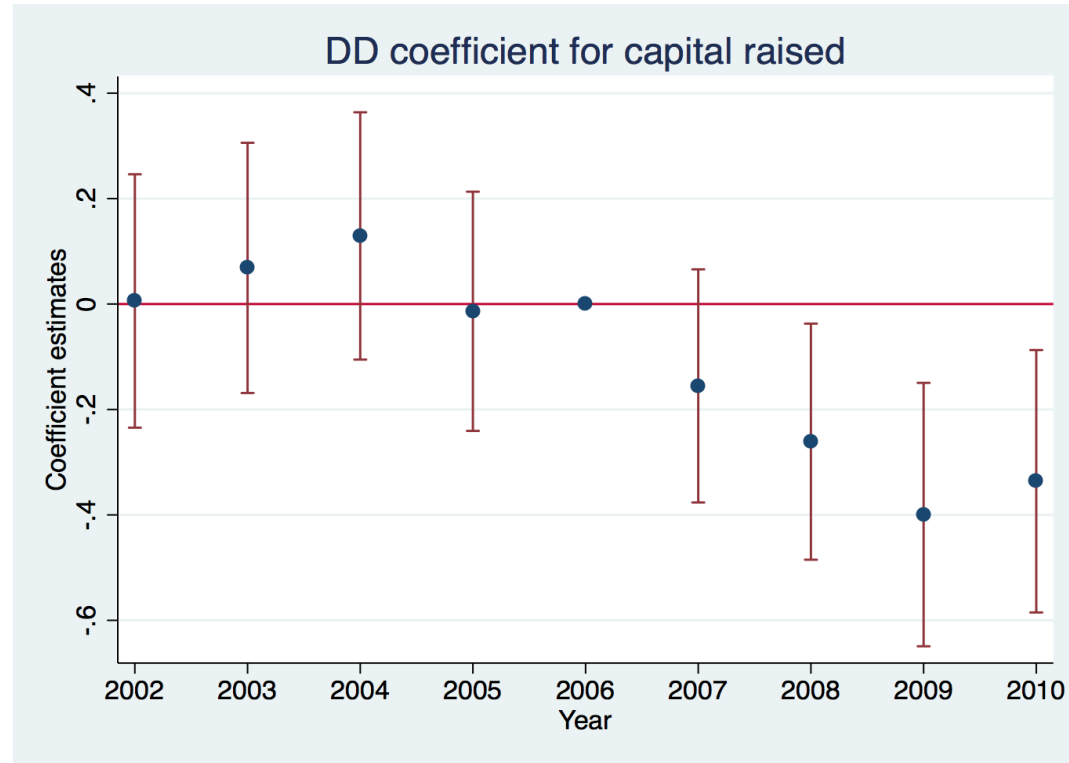


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 - Cloud computing's emergence in 2006 (AWS) → build software products w/o significant upfront investments → raise less capital (Ewens et al. 2018)
 - Mark Andreessen:
 - *“In the 90s, if you wanted to build an Internet company, you needed to buy Sun servers, Cisco networking gear, Oracle databases, and EMC storage systems... and those companies would charge you a ton of money even just to get up and running. The new startups today, they don't buy any of that stuff...”*
 - Mark Suster:
 - *“Amazon in turn led to the formation of an earlier stage of venture capital now led by “micro VCs” who typically invest \$250-500k in companies rather than the \$5-7 million that VCs used to invest.”*

Changes affecting startups and founders

Introduction of the “cloud” as a shock to the cost of experimentation

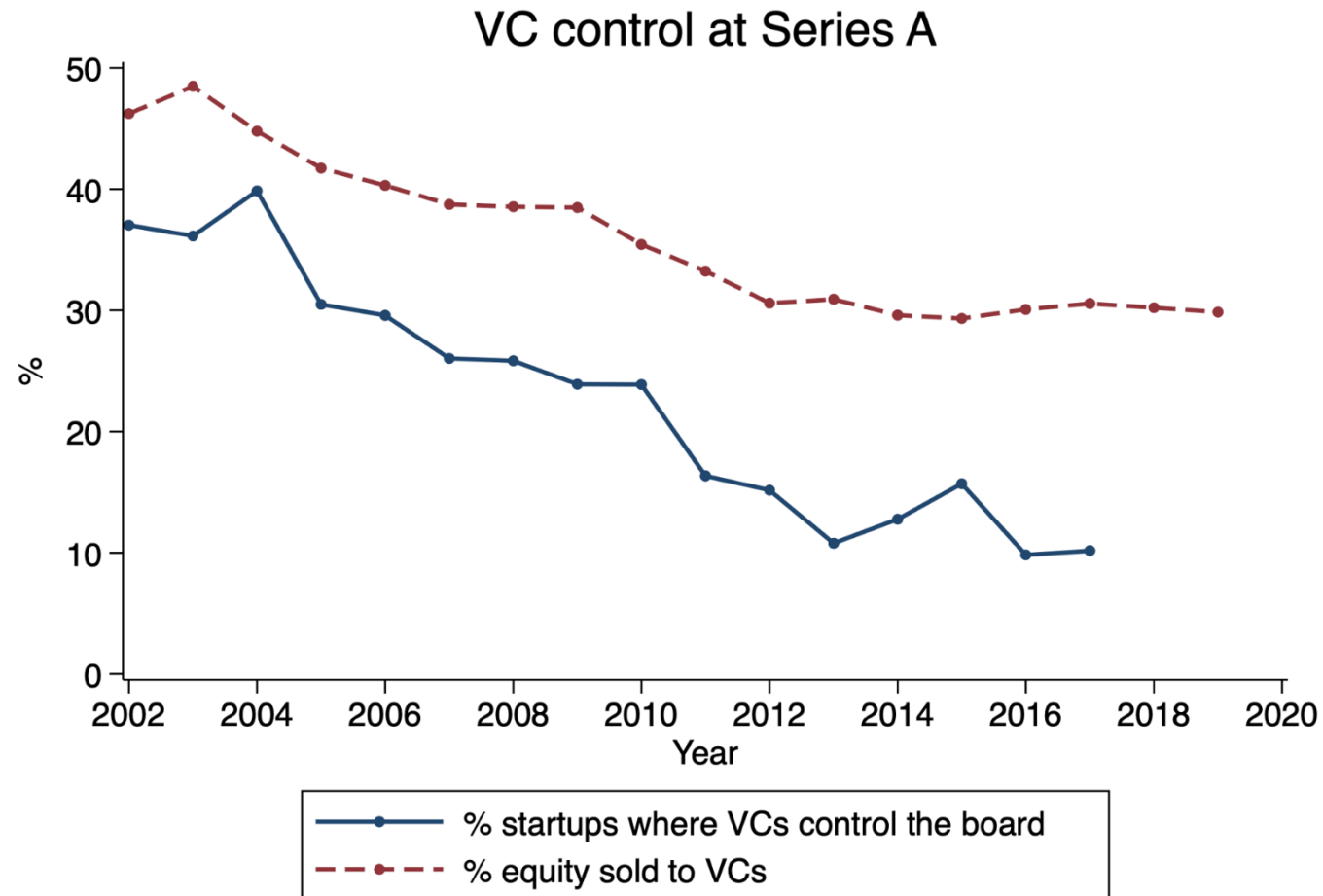


15-29% reduction in capital raised in first round financings

Connecting the dots

- Regulatory changes and investor-level changes have increased the **supply** of private capital
- Technological changes have decreased startups' **demand** for capital, particularly in early stages, when capital is relatively most expensive
- When startups reach age at which they could go public, founders retain more control of their firms → more influence on exit decisions
 - Founders often prefer staying private (less scrutiny, more control; Brau & Fawcett 2006)

Founders have more bargaining power to make exit decisions, which allows them to delay exits



More founder equity → less likely to exit

Do increases in founder equity lower probability of exit? (yes)

- Ewens & Farre-Mensa (2020): exploit variation in founder equity stemming from exogenous shifts in VC supply when startup founded
 - IV: State pension fund assets $\times$ % state official in board of trustees (“home bias”)
 - More abundant VC supply → Founders get to keep more equity
- Instrumented increases in founder equity decrease prob. that a startup exits (IPO or M&A) w/in 7 yrs of first VC

Do these changes work for everybody?

- Problem for founders and employees: lack of liquidity (“paper millionaires”)
 - Growth in secondaries might help alleviate
- Problem for early investors: exits are delayed, possibly beyond the end of their funds’ lifecycle
 - Can secondaries solve this? (e.g., Series A VC sells to Series E VC/PE)
 - Unclear: Adverse selection concerns stronger than w/ founders/employees
 - Evergreen funds, changes to funds lifecycles?

Regulators' response to this new environment

Spillovers in capital market regulations

Regulatory tug-of-war: JOBS Act as a case study

Public market deregulation

Emerging Growth Companies (Title I)

[General Applicability of Title I](#)

[FAQ: Confidential Submission Process for Emerging Growth Companies](#)

[Research Analysts and Underwriters](#)

Easier to go and be public as a “small” firm

Private capital market deregulation

Access to Capital for Job Creators (Title II)

[Compliance Guide: Eliminating the Prohibition Against General Solicitation](#)

[FAQs: Exemption from Broker-Dealer Registration in Title II of the JOBS Act](#)

[Voluntary Submission of Rule 506\(c\) Materials](#)

Crowdfunding (Title III)

[Compliance Guide for Issuers](#)

[Compliance and Disclosure Interpretations for Issuers](#)

[Funding Portal Registration Guide](#)

[Compliance Guide for Intermediaries](#)

[Frequently Asked Questions Regarding Regulation Crowdfunding and Intermediary Requirements](#)

Lower private capital raising frictions

Expand retail investors access to VC/PE

SEC's Asset Management Advisory Committee "Final Report and Recommendations for Private Investments":

- Asset management growth in self-directed retirement
- Decline in the # of investment opportunities (i.e., "where are the small caps"?)
- → allow retirement accounts or individuals to invest in PE

VC/PE could possibly provide (Brown et al., 2020)

- i. higher expected returns
- ii. diversification benefits

Expand investors' access to VC/PE

Regulation Crowdfunding

Regulation Crowdfunding enables eligible companies to offer and sell securities through crowdfunding. The rules:

SEC Modernizes the Accredited Investor Definition

FOR IMMEDIATE RELEASE
2020-191

“[W]e view the supply side for retail investors as being less diversified than 15 years ago.”
- Asset Management Advisory Committee
Final Report and Recommendations for Private Investments

Should defined contribution plans
include private equity investments?¹

December 12, 2020

Brown et. al. (2020)

REGULATION

House Bill Would Expand Access to Private-Equity Funds

A bipartisan proposal comes amid a regulatory push to allow private funds to raise money from ordinary investors

By [Chris Cumming](#)

July 9, 2021 6:08 pm ET

Regulatory challenges of increased investor access

- To protect retail investors in private mkts, may need to regulate private firms more like public ones:
 - E.g., shrink disclosure gap btw/ public and private firms

→ **Private firms may lose some of their key advantages**

SEC Proposes to Enhance Private Fund Investor Protection

FOR IMMEDIATE RELEASE
2022-19

Washington D.C., Feb. 9, 2022 — The Securities and Exchange Commission today voted to propose new rules and amendments under the Investment Advisers Act of 1940 (Advisers Act) to enhance the regulation of private fund advisers and to protect private fund investors by increasing transparency, competition, and efficiency in the \$18-trillion marketplace.

Sanctions Spur Calls to Make Private Equity Vet Investors

Lawmakers concerned about Russian and Chinese money in U.S. private funds want to bring investors' identities to light

SEC Pushes for More Transparency From Private Companies

'Unicorn' firms have a huge impact and 'absolutely no visibility' for regulators, says SEC Commissioner Allison Lee

Open research and policy questions

- Connection btw startup higher valuations and + founder-friendly contracts, and returns earned by VC/PE asset class (Harris et al. 2020)?
- How do the changes in the late-stage financing market impact the traditional VC investing model?
- Real consequences of decline in listed firms: e.g., less information on prices and investment opportunities → less efficient investment?
- How can other countries foster a thriving entrepreneurial finance market, particularly for late-stage startups?
 - “[EU]later-stage financing in particular remains restricted” (EC 2018)

Thank you!

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mewens@caltech.edu](https://privatepublicmkts.com/mewens@caltech.edu)